MS&E 228 (Winter 2026) — Lecture 17 Notes

Difference-in-Differences and Regression Discontinuity Designs

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March 15, 2026

Readings. *Applied Causal Inference Powered by ML and AI*, Chapters 16–17. Key papers: Card & Krueger (1994), Demirer et al. (2024), Carpenter & Dobkin (2009), Almond et al. (2010), Calonico, Cattaneo, & Titiunik (2014), Card, Dobkin, & Maestas (2009), Van der Klaauw (2002).

1 Introduction: Beyond Graphical Criteria

This chapter introduces two of the most widely used techniques in applied causal inference, particularly in applied econometrics, the social sciences, and healthcare: **Difference-in-Differences (DiD)** and **Regression Discontinuity (RDD)**. These methods provide powerful strategies for addressing unobserved confounding, but they operate on a fundamentally different set of principles than the graph-based methods we have studied so far.

1.1 Recap: Three DAG-Based Strategies

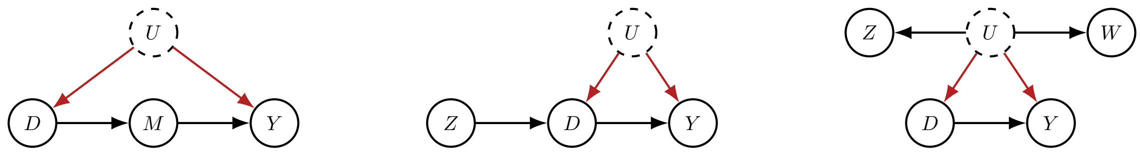
In the last few lectures, we explored three primary strategies for identification in the presence of unobserved confounders. Each of these methods relied on identifying a specific graphical structure in the underlying causal DAG. If the DAG of a given problem matched one of these templates, we could apply the corresponding identification formula:

1. **The Front-Door Criterion:** This strategy required the existence of a *complete mediator*—a variable M that lies on all causal paths from the treatment D to the outcome Y, and which is itself unconfounded with Y given D.

2. **Instrumental Variables (IV):** This approach required an *instrument* Z that satisfies three conditions: it is relevant (it affects the treatment D), it satisfies the exclusion restriction (it affects the outcome Y only through D) and it is conditionally exogenous (we can estimate effects of intervening on the instrument, by conditioning).

3. **Proximal Causal Inference:** This method required two proxy variables: a *treatment-inducing proxy* and an *outcome-inducing proxy*, each correlated with the unobserved confounder through different channels.

At the end of the previous lecture, we compared these three methods side-by-side. Visually, each corresponds to a representative graph that you should keep in mind. If you are faced with a new causal inference problem, you can sketch a causal graph for it, and if it resembles any of these prototypical graphs, you can apply the corresponding method.



(a) Front-Door (b) Instrumental Variable (c) Proximal Inference

It is important to note that in all of these methods, the distinction between *identification* and *estimation* was clear. The graphical structure gave us identification. Assumptions like partial linearity, which we used for certain estimators, were made primarily for *estimation* purposes—they helped us construct practical estimators—but they were not required for identification itself.

1.2 Identification from DAGs vs. Functional Form

The approaches we cover today are qualitatively very different. The assumptions that enable identification are *not* based on identifying a particular type of graph structure in the DAG. Instead, they are based on imposing further restrictions on the *functional form* of the structural equations. This information is not contained in the graph itself; it represents a stronger set of assumptions about the data-generating process than what parental relationships alone can encode. You cannot simply "read off" the identifying assumptions from a DAG the way you can with the front-door criterion or instrumental variables.

The goals for this chapter are to:

1. Introduce the core ideas behind **Difference-in-Differences (DiD)** and how it uses panel data (repeated observations over time) to control for time-invariant unobserved confounders.

2. Introduce **Regression Discontinuity Designs (RDD)**, which exploit thresholds or cutoffs in treatment assignment rules to estimate local causal effects.

3. Understand how both methods hinge on **functional form assumptions** (parallel trends for DiD, smoothness for RDD) that are distinct from graphical criteria.

4. Explore various **recent empirical applications** of these methods.

2 Difference-in-Differences (DiD)

2.1 Motivating Example: A/B Testing with Confounding

Let us introduce the DiD setting with a common scenario from the tech industry. Suppose that Microsoft wants to release a new feature in its Office suite, such as a generative AI assistant (GitHub

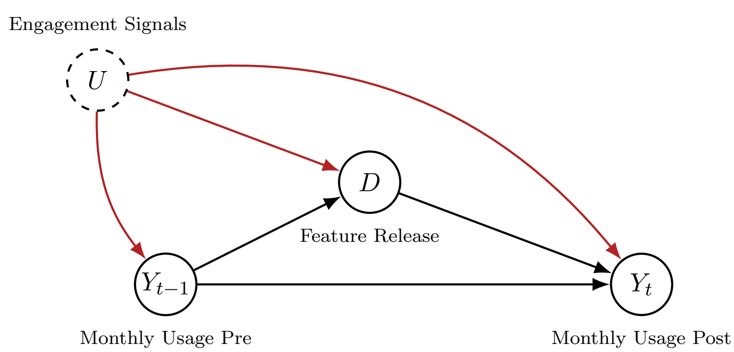
Copilot) for PowerPoint. Software companies often release new features in waves, starting with a small group of users to gather feedback before a wider rollout. Typically, this first wave consists of the most engaged or trusted users—those who use the software frequently and are likely to provide valuable insights. The company looks at a rich set of internal engagement signals, which we can call U, and based on these signals, it characterizes who is a first-wave user and who is a second-wave user.

This is not a hypothetical story. There are many recent papers by Microsoft researchers using internal Microsoft data to understand the effect of Copilot on productivity (see, e.g., Dillon et al., 2025). As a researcher at Microsoft, you would receive a dataset of, say, 10,000 users, their monthly activity levels on PowerPoint, and information about when the feature was released to the first wave. However, when you receive this data, you typically do not get all of the internal information that the company used to decide who to treat. The company might share a subset of that information, but the full set of engagement signals U remains unobserved.

Based on this data, our goal is to understand the causal effect of the new feature on user activity. We want to understand not just what happened after the release, but the *causal* effect of the feature on the level of activity for the users who were treated, even though treatment assignment was based on unobserved confounders.

2.2 The Causal Structure

Let us draw the causal graph for this problem. Suppose the feature was released in March. Let Y_{t-1} denote the pre-treatment usage (e.g., February) and Y_t denote the post-treatment usage (March). The question of who was treated by the release is driven by the unobserved engagement signals U. Potentially, one of those signals could even be the monthly usage itself.



Discussion: Can We Apply Previous Methods?

Looking at this graph, can we identify the causal effect of D on Y_t using any of the techniques we have discussed so far?

Potentially: "We could condition on the pre-treatment variable Y_{t-1} and then do some sensitivity analysis."

That is one approach, but does not yield identification of the effect. Conditioning on Y_{t-1} alone would not block all backdoor paths. Do any of the prototypical graphs for the other strategies apply? Do we have a front-door variable, a complete mediator, or an instrument? Do we have pre-treatment proxies or outcome proxies? No. Just from the graph, there is nothing we can directly use. We need a new kind of assumption.

2.3 The Core Idea: The Parallel Trends Assumption

At this point, we move beyond the graph and start making assumptions about the structural equations themselves. The graph is a representation of the structural equations, which so far have been very flexible—we have made no assumptions about what they look like. Now, we will introduce assumptions that implicitly impose restrictions on these structural equations.

Here is the core assumption. We are willing to believe that the engagement signals (U) that the company used to decide who to treat are indicative of *usage levels*. In other words, people who engage more will have higher levels of usage. These signals therefore act as an unobserved confounder: if we were to simply compare the usage of the treated group versus the control group, we would find a significant difference. However, we are willing to believe that these signals are *not* indicative of how usage will *grow* over time in the absence of treatment.

We can say that highly engaged users will consistently have very high levels of usage. But we do not expect that, for instance, between February and March, these highly engaged users will increase their usage by an additional 20 hours on average, while low-engaged users will only increase their usage by one hour on average. If there is some growth—perhaps due to more people using Office in general, creating a time trend—that average growth will be the same for both the treated and the control populations.

The Parallel Trends Assumption

Let $Y_t(0)$ be the potential outcome (e.g., usage) at time t in the absence of treatment. The parallel trends assumption states that the expected change in the potential outcome under control is the same for both the treated and control groups. For a pre-treatment period t' and a post-treatment period t:

$$\mathbb{E}[Y_t(0) - Y_{t'}(0) \mid D = 1] = \mathbb{E}[Y_t(0) - Y_{t'}(0) \mid D = 0]$$

This is a form of **one-sided ignorability**, but it applies to the *change* in outcomes,

$$\Delta Y_{t,t'} = Y_t - Y_{t'},$$

rather than the levels. Note that we are leveraging the fact that we observe not just one outcome for a user, but a *time series* of outcomes—both pre-treatment and post-treatment observations for the same outcome.

2.4 From Parallel Trends to Identification

Since we have one-sided ignorability for the change in outcomes between a post-treatment period and a pre-treatment period, we can ask: what quantity can we estimate? Is it the ATE, the ATT, the LATE, or the CATE?

Question: What Can We Identify?

Given one-sided (unconditional) ignorability for ΔY, what causal quantity can we identify? **Answer:** The **Average Treatment Effect on the Treated (ATT)** for the change in outcomes. Since we have one-sided ignorability (the change in the potential outcome under control is mean-independent of the treatment assignment), we can identify the ATT, not the ATE.

For any post- and pre-treatment period, we can define the ATT on the change in outcomes:

$$\text{ATT-on-Change} = \mathbb{E}[\Delta Y_{t,t'}(1) - \Delta Y_{t,t'}(0) \mid D = 1]$$

This is identified by the difference in the average observed change between the two groups:

$$\text{ATT-on-Change} = \mathbb{E}[\Delta Y_{t,t'} \mid D = 1] - \mathbb{E}[\Delta Y_{t,t'} \mid D = 0]$$

The first term is the average change in the treated population, and the second is the average change in the control population. The parallel trends assumption allows us to replace the average unobservable counterfactual change $\Delta Y_{t,t'}(0)$ for the treated group with the average observed change in the control group.

2.5 From ATT-on-Change to ATT-at-t

We have estimated the effect on the *change* in outcomes. But what do we really want to estimate? We set out to estimate some form of effect on the outcome at some post-treatment period t. For instance, the average effect on the treated at period t, which is:

$$\text{ATT-at-}t = \mathbb{E}[Y_t(1) - Y_t(0) \mid D = 1]$$

We have only managed to identify the ATT on the change in outcomes. How do these two quantities relate? Let us expand the ATT-on-Change:

$$\begin{aligned}\text{ATT-on-Change} &= \mathbb{E}\big[(Y_t(1) - Y_{t'}(1)) - (Y_t(0) - Y_{t'}(0)) \mid D = 1\big] \\ &= \underbrace{\mathbb{E}[Y_t(1) - Y_t(0) \mid D = 1]}_{\text{ATT-at-}t} - \underbrace{\mathbb{E}[Y_{t'}(1) - Y_{t'}(0) \mid D = 1]}_{\text{Effect in pre-treatment period}}\end{aligned}$$

Question: Relating the Two ATTs

When does ATT-on-Change equal ATT-at-t?

Answer: Looking at the expansion above, we need the second term—the effect of the treatment on the pre-treatment outcome—to be zero. This requires an additional assumption.

2.6 The No Anticipation Assumption

To isolate the average treatment effect at a specific post-treatment period, we need the effect of the treatment on any pre-treatment period to be zero. This is essentially an "arrow of time" assumption: if we treat a unit in March, that treatment has no effect on its outcomes in February.

The No Anticipation Assumption

The **no anticipation assumption** states that a treatment implemented at time $t_{release}$ has no effect on outcomes in any pre-treatment period $t' < t_{release}$. Formally:

$$Y_{t'}(1) = Y_{t'}(0) \quad \text{for all } t' < t_{release}$$

This implies that

$$\mathbb{E}[Y_{t'}(1) - Y_{t'}(0) \mid D = 1] = 0.$$

This assumption is very natural, but it can be violated in practice in subtle ways. It is always good to reason qualitatively about why it is not violated in the setting you are working with. Consider the following examples:

When No Anticipation Fails

- **Announced treatments:** If Microsoft announced that the new feature would be given to the most active users, people might increase their activity in anticipation of the release to ensure they get treated. In a public policy context context, if the European Union announces in January that the GDPR policy will be enforced in March, then the months between January and March might be affected by the announcement. If our goal is to estimate the effect of GDPR on firm productivity, then the no anticipation assumption might be violated for these months between announcement of the treatment and the treatment itself.

- **Gamification and badges:** On online platforms, badges or rewards (e.g., credits) are sometimes offered for users who achieve certain engagement milestones. On Xbox, playing a certain number of hours might earn a "Pro" badge; on Stack Overflow, a certain number of posts makes you a "golden user." If we then want to measure the effect of that badge on downstream activity, the no anticipation assumption is violated because people change their activity specifically to earn the badge.

- **Algorithmic assignment rules:** More generally, if the rule for treatment assignment is announced beforehand, people may change their behavior prior to the treatment so as to be treated.

However, in many cases—such as sudden policy decisions or product releases that are implemented immediately without prior announcement—the treatment should not affect prior periods, and the no anticipation assumption is more likely to hold. In cases when there was a pre-treatment announcement, we could potentially avoid these violations if we use as pre-treatment period some period before the announcement itself.

2.7 The Difference-in-Differences Estimator

Under both the parallel trends and no anticipation assumptions, we have

$$\text{ATT-at-}t = \text{ATT-on-Change}.$$

This gives us the famous Difference-in-Differences identification formula.

The Difference-in-Differences Identification and Estimation

The ATT at a post-treatment period t is identified by the difference in the change in outcomes between the treated and control groups:

$$\text{ATT-at-}t = \underbrace{\left(\mathbb{E}[Y_t \mid D=1] - \mathbb{E}[Y_{t'} \mid D=1]\right)}_{\text{Expected Change in Treated}} - \underbrace{\left(\mathbb{E}[Y_t \mid D=0] - \mathbb{E}[Y_{t'} \mid D=0]\right)}_{\text{Expected Change in Control}}$$

This is the "difference in differences": we calculate the difference in outcomes over time for the treated group, we calculate the difference in outcomes over time for the control group, and then we take the difference of these two differences as our final quantity.
When we have finite samples, we can simply replace the expectations in the formula above with empirical averages, leading to the famous difference-in-differences estimator:

$$\hat{\theta}_t = \underbrace{\left(\frac{1}{n_1}\sum_{i:D_i=1}(Y_{t,i} - Y_{t',i})\right)}_{\text{Average Change in Treated}} - \underbrace{\left(\frac{1}{n_0}\sum_{i:D_i=0}(Y_{t,i} - Y_{t',i})\right)}_{\text{Average Change in Control}}$$

where n_1, n_0 is the number of treated and control samples, respectively.

2.8 Placebo Tests

The parallel trends assumption is strong and untestable directly from the data. Because it is a strong assumption, we can use some tests that could potentially reject it. One such test is a **placebo test**. In a placebo test, we look at two *pre-treatment* periods t, t' and examine the difference between the control and the treated populations. Since no treatment occurred during this time, we should expect to get a zero effect. If we apply the DiD estimator for two pre-treatment periods and find a non-zero effect, it could mean one of two things:

1. The **parallel trends assumption** does not hold, or
2. There is an **anticipation effect** in the treated population.

A non-zero placebo test is a warning sign that the DiD analysis may not be valid. Typically, we will be using a the same "pre-treatment period" anchor t' for both our target effects and our placebo tests. For instance, we could be using the last period before treatment as $t' = t_{release} - 1$ for estimating the effect of all post treatment periods. Then we should be running our placebo test with the same t' and with t being some other earlier pre-treatment periods, e.g. $t_{release} - 2$.

2.9 Empirical Example: GDPR and Firm Production

To ground these ideas in a real-world application, let us discuss a recent paper by Demirer et al. (2024) on the economic consequences of the General Data Protection Regulation (GDPR).

Widespread Empirical Use of DiD

If you explore the difference-in-differences literature, you will find that there have been on the order of *thousands* of papers, if not more, that have applied the DiD approach in the field of empirical causal inference. This fact implies two things: the method is widely used, but it is also *widely abused*. It is simply not plausible that the parallel trends assumption held true in all of the settings where it has been applied. The main reason for its wide applicability is its ease of use and the accessibility of data suitable for the analysis. As long as you have a dataset with repeated observations of the same outcome, you can blindly apply DiD without making a strong argument about the data-generating process. For this reason, it is often abused. Because the argument is that it makes weaker assumptions than exogeneity, researchers often claim they have implicitly controlled for many factors simply by doing a DiD. That said, the Demirer et al. paper is a good example of a careful application.

The study sought to understand the effect of GDPR, introduced in the European Union around June 2018, on tech investment, firm production, and the data usage of firms.

- **Treatment:** GDPR regulation.
- **Treated Group:** Firms in the European Union.
- **Control Group:** Firms in the United States with no EU presence (because US firms with some EU presence were also affected by GDPR).
- **Data:** Firm-level cloud usage data scraped from a global provider of firm-level usage data.

What does the parallel trends assumption mean in this context? It essentially means that we believe that, while companies in the US might tend to use more cloud services than companies in Europe on average for various reasons, the *change* in cloud usage between, say, May and July of 2018 for US companies should, on average, be the same as the change in cloud usage for EU companies *would have been* if GDPR had not been implemented. That is the core of the parallel trends assumption here.

2.9.1 Demirer et al. (2024): Key Findings

The paper produced plots showing the DiD estimators applied to each individual post-treatment period compared to a pre-treatment period. There was a clear negative effect on cloud and data usage from GDPR. The authors estimated the ATT and a 95% confidence interval for every post-treatment period, producing a time series of estimated effects.[1]

Crucially, the paper also included **placebo tests** for the pre-treatment periods. The month right before GDPR was introduced served as the anchor period, and the pre-treatment effects were

[1] The analysis in Demirer et al (2024) controls for some firm characteristics too; hence the identification argument and analysis is closer to the conditional parallel trends assumption that we introduce in the next section.

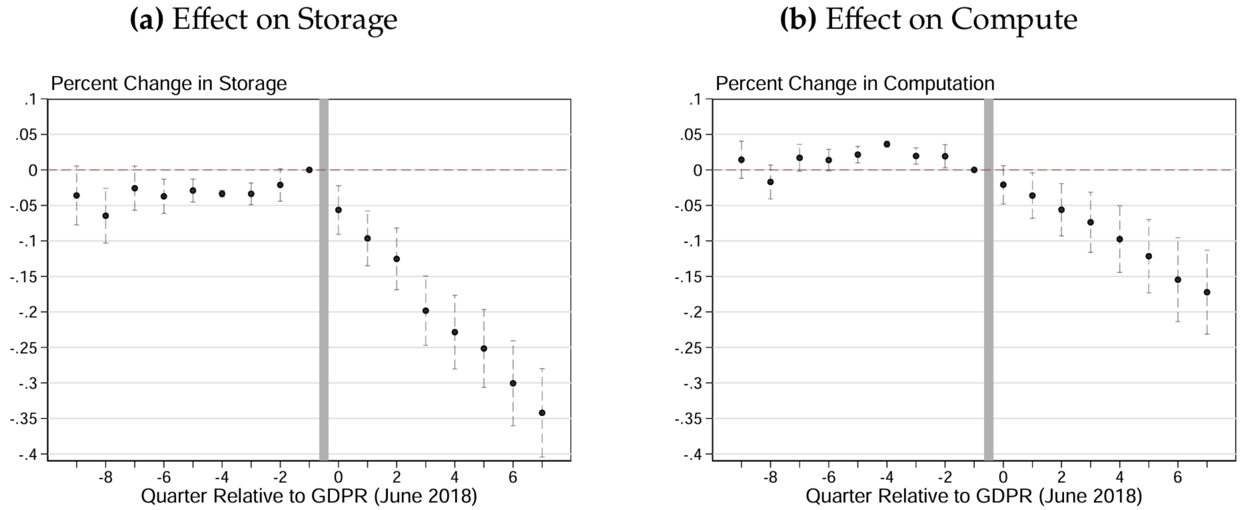


Figure 2: Empirical results of Demirer et al. (2024).

calculated between this anchor and each earlier period. Based on our discussion, these pre-treatment effects are placebo tests, and they should all be statistically zero.

In the results, there were some violations of the assumptions—some pre-treatment effects were statistically non-zero. However, the magnitude of these violations was not large compared to the post-treatment effects. One could use the magnitude of the pre-treatment violation as a guide to how much the omitted variable bias could be, and then impose bounds on the actual effects. Given how small the magnitude of the violation was, it should not invalidate the main finding. These are the kinds of plots that you typically include when you write a paper that uses a DiD approach.

2.10 Conditional Parallel Trends

The simple (unconditional) parallel trends assumption can be made more plausible by conditioning on observed covariates X. It is plausible that in the GDPR scenario, the composition of firms in the US is very different from the composition of firms in the European Union. For instance, we likely have more tech firms in the US than in the EU, or differences in other sectors. Moreover, it is plausible that different sectors grow differently in terms of cloud usage—you might expect that tech companies grow faster in terms of cloud usage than other sectors.

This means that the unconditional parallel trends assumption should not really hold in this scenario. If the composition of firms is different in the two groups in terms of observable characteristics, and if these observable characteristics lead to different average changes, then we should expect that the overall average change in the control group should be different than the overall average change in the treated group.

A better assumption is to condition on all observables. The **conditional parallel trends assumption** says that once we condition on all of the observable characteristics of a firm—which could be sector, size, prior cloud usage, and so on—we should not expect to find discrepancies in the baseline growth trajectory between treated and controlled firms.

The Conditional Parallel Trends Assumption

For any value x of the covariates X, the expected change in the potential outcome under control for units with characteristics x, is the same for both the treated and control groups:

$$\mathbb{E}[\Delta Y_{t,t'}(0) \mid D = 1, X = x] = \mathbb{E}[\Delta Y_{t,t'}(0) \mid D = 0, X = x]$$

This is one-sided **conditional** ignorability for the change in outcomes.

2.11 Estimation with Conditional Parallel Trends

Under the conditional parallel trends assumption, we have one-sided *conditional* ignorability for the change in outcomes. This allows us to identify the ATT using methods for conditional ignorability. However, it is not as simple as using a basic difference-in-means estimator, because we are now dealing with *conditional* exogeneity. We need to use all the fancy identification formulas that we already know for the ATT (e.g., g-formula, IPW formula, DR formula). In this case, for the treated group—in the GDPR example, the EU firms—we can simply calculate their average growth. But for the control group (the US firms), we cannot just take their average growth. Instead, we need to calculate the average over a *re-weighted* composition of US firms to make them look like the EU firms in terms of their observable characteristics.

The **g-formula for the ATT** accomplishes this by first going to the US firms and estimating a growth model as a function of X. This model determines the expected growth for US firms given a set of firm characteristics X, which we can define as a function $g_0(X) = \mathbb{E}[\Delta Y_{t,t'} \mid D = 0, X]$. Next, the formula instructs us to go to each of the EU firms. For each firm, we take its observed growth and subtract the baseline growth that we learned from the US population for a firm with similar characteristics. This baseline represents the growth we would have expected this specific company to have if GDPR had not been released. Finally, we take the average of this difference across all the EU firms.

That is the g-formula for the ATT, which when combined with the no-anticipation assumption gives us the g-formula for the DiD setting under *conditional* parallel trends:

$$\text{ATT-at-}t = \text{ATT-on-Change} = \mathbb{E}[\Delta Y_{t,t'} - g_0(X) \mid D = 1], \qquad g_0(X) = \mathbb{E}[\Delta Y_{t,t'} \mid D = 0, X]$$

2.12 Doubly Robust Estimation for DiD

If the firm characteristics are rich enough—meaning you have a variety of covariates—you will likely want to estimate the growth model using a generic machine learning approach. As we have discussed many times in this course, when using machine learning for nuisance estimation, the recommended approach is to use *insensitive formulas* and *cross-fitting*. In this case, this corresponds to the **Doubly Robust (DR) estimator for the ATT**.

We have already seen the doubly robust estimator for the ATT; you implemented it in one of your early homeworks. Now, you just need to re-implement the exact same thing. The only difference is that instead of passing the outcome at period t to the estimator, you pass the *change in outcomes* between period t and some pre-treatment period t'. If you do that, you will be estimating a doubly robust ATT for that post-treatment period t. You can do this for any post-treatment period t, and you will get a plot like the one in the Demirer et al. paper—a doubly robust analog of that plot.

Note that the Demirer et al. paper did not use the doubly robust estimator; it used a more classical approach that people tend to use, called two-way fixed effects with some extra controls added in a linear manner (primarily due to the ease of implementation of the estimation procedure via OLS). In this class, we are only going to learn the doubly robust estimator for this setting. The DR approach is preferable as it does not rely on any linearity or partial linearity assumptions.

Reminder: The Doubly Robust ATT Estimator

Recall the DR ATT formula from Lecture 4. For a binary treatment D, outcome $\tilde{Y}$ (which in the DiD setting is $\Delta Y = Y_t - Y_{t'}$), outcome regression on controls $g_0(X) = \mathbb{E}[\tilde{Y} \mid D = 0, X]$, and propensity score $p_0(X) = \Pr(D = 1 \mid X)$, the DR ATT formula is

$$\text{ATT} = \frac{\mathbb{E}\Big[D\,(\tilde{Y} - g_0(X)) - (1 - D)\,\frac{p_0(X)}{1-p_0(X)}\,(\tilde{Y} - g_0(X))\Big]}{\mathbb{E}[D]}.$$

The empirical plug-in analogue is now the ratio of two sample averages:

$$\widehat{\text{ATT}} = \frac{\mathbb{E}_n\Big[D\,(\tilde{Y} - \hat{g}(X)) - (1 - D)\,\frac{\hat{p}(X)}{1-\hat{p}(X)}\,(\tilde{Y} - \hat{g}(X))\Big]}{\mathbb{E}_n[D]}.$$

If the nuisance functions are trained with cross-fitting, we have strict one-sided overlap, i.e., $\Pr(D = 1 \mid X) < 1-\epsilon$ a.s., and the product rate condition holds $\sqrt{n}\,\text{RMSE}(\hat{g})\,\text{RMSE}(\hat{p}) \to 0$, then the estimate is asymptotically normal

$$\sqrt{n}(\widehat{\text{ATT}} - \text{ATT}) \Rightarrow \mathcal{N}(0, V_{\text{ATT}}), \qquad V_{\text{ATT}} = \text{Var}(\phi_0(Z)),$$

where the function $\phi_0(Z)$ is defined as:

$$\phi_0(Z) = \frac{m_0(Z) - \text{ATT} \cdot D}{\mathbb{E}[D]} \qquad m_0(Z) := D\,(Y - g_0(X)) - (1 - D)\,\frac{p_0(X)}{1 - p_0(X)}\,(Y - g_0(X)).$$

The standard error $\sqrt{V_{\text{ATT}}/n}$ can be estimated in a plugin manner and used for confidence interval construction.

The pseudocode below provides a detailed implementation of the DR-DiD estimator, mirroring the code from our previous lectures.

Python: Doubly Robust DiD Estimator

```python
import numpy as np
import pandas as pd
from sklearn.model_selection import KFold
from sklearn.ensemble import RandomForestRegressor, RandomForestClassifier
from sklearn.base import clone

# 1. Define the change in outcome
delta_Y = data["Y_post"] - data["Y_pre"]
X = data[["X1", "X2"]]  # Covariates (sector, size, etc.)
D = data["D"].values      # Treatment indicator (1=EU, 0=US)
n = len(data)

# 2. Cross-fitting: estimate nuisance functions
cv = KFold(n_splits=5, shuffle=True, random_state=123)
g0hat = np.zeros(n)  # outcome regression on controls
phat  = np.zeros(n)  # propensity score

model_g = RandomForestRegressor(n_estimators=200, min_samples_leaf=10)
model_p = RandomForestClassifier(n_estimators=200, min_samples_leaf=10)

for train_idx, test_idx in cv.split(X):
    # Fit outcome model g(X) = E[delta_Y | D=0, X] on controls only
    ctrl_mask = (D[train_idx] == 0)
    g = clone(model_g).fit(X.iloc[train_idx][ctrl_mask],
                           delta_Y.iloc[train_idx][ctrl_mask])
    g0hat[test_idx] = g.predict(X.iloc[test_idx])

    # Fit propensity score model p(X) = P(D=1 | X)
    p = clone(model_p).fit(X.iloc[train_idx], D[train_idx])
    phat[test_idx] = p.predict_proba(X.iloc[test_idx])[:, 1]

# 3. Compute the DR scores for ATT on the change
ipw_weight = phat / (1 - phat)
m = D * (delta_Y.values - g0hat) \
    - (1 - D) * ipw_weight * (delta_Y.values - g0hat)

# 4. Compute DiD estimate and 95% CI
did_att = m.mean() / D.mean()
phi = (m - did_att * D) / D.mean()
se = np.sqrt(np.var(phi) / n)
ci_lo, ci_hi = did_att - 1.96 * se, did_att + 1.96 * se

print(f"DR-DiD ATT Estimate: {did_att:.4f}")
print(f"95% CI: [{ci_lo:.4f}, {ci_hi:.4f}]")
```

Discussion: Cross-Fitting Details

Question: "Should we use K-fold or group K-fold here?"
Response: We do not need to use group K-fold here because you only have one sample for every unit. For each unit i, you have a pre- and post-treatment period, which constitutes your sample. Since you are repeating the exercise for every pre- and post-treatment period pair, you can just do a simple K-fold because you are splitting the units, not the times. For every unit, you have just one pre- and post-period, so you just run K-fold across the units. You do not need to do group K-fold.

Discussion: Continuous Treatments

Question: "What about continuous treatments? What does DiD mean in that context?"
Response: With different levels of treatment, the methodology becomes more nuanced. Are we only assuming parallel trends for the zero-treatment group? There are ways to extend the methodology for continuous treatments. In summary, with continuous treatments, you would take the difference between post and pre-treatment outcomes and run the Double ML algorithm with the difference as the outcome. For instance, define 0 to be some baseline treatment and assume that $\Delta Y_{t,t'}(0) \perp\!\!\!\perp D \mid X$. Under this setting, we can still write:

$$\mathbb{E}[Y_t(D) - Y_t(0) \mid D] = \mathbb{E}[\Delta Y_{t,t'}(D) - \Delta Y_{t,t'}(0) \mid D] = \mathbb{E}[\Delta Y_{t,t'} - g_0(X) \mid D]$$

If we further assume that there is a group of units that received zero treatment, i.e. there is a group of samples with $D = 0$, then we can estimate $g_0(X) = \mathbb{E}[\Delta Y_{t,t'} \mid D = 0, X]$, by running a regression on that group. Finally, if we assume a partially linear model on the growth:

$$\mathbb{E}[\Delta Y_{t,t'}(D) - g_0(X) \mid D, X] = \theta' \phi(D)$$

for some known feature map $\phi(D)$ with $\phi(0) = 0$, then we can estimate the parameters θ by running the DoubleML algorithm with the difference as the outcome. Then we can estimate an ATT dose response curve, i.e. the average effect at period t, of receiving dose d, for people that were treated with dose d as:

$$\mathbb{E}[Y_t(d) - Y_t(0) \mid D = d] = \theta' \phi(d)$$

2.13 Placebo Tests under Conditional Parallel Trends

The same idea of placebo tests applies under conditional parallel trends, but now we use the doubly robust estimator instead of simple differences in means. We apply the DR-DiD estimator to two different pre-treatment periods and check whether the estimated effect is statistically zero.

Discussion: Choosing the Pre-Treatment Period

Question: "How do we choose the pre-treatment time interval? Does it matter if we test many different ones?"

Response: Typically, for the sake of cleanliness, there are three main strategies:

1. **Use the last pre-treatment period** as the anchor.
2. **Use a uniform average** of all the pre-treatment periods.
3. **Use a weighted average**, putting more weight on more recent periods. There are strategies that define the weights to minimize the variance (Egami and Yamauchi, '23).

In principle, you could repeat this exercise with two or three pre-treatment periods and see if the effects differ. If the estimated effects are statistically different, that is also problematic. You can also run the same placebo tests using the doubly robust estimator for two different pre-treatment periods and check if the confidence intervals from the DR ATT estimator overlap. Typically, as done in the Demirer et al. paper, you would have the last pre-treatment period as your anchor, and then measure every other pre-treatment period against that anchor.

3 Regression Discontinuity Designs (RDD)

3.1 The Core Idea: Treatment at a Threshold

The second major strategy we explore is the **Regression Discontinuity Design (RDD)**. This method is applicable in many settings where a treatment is administered based on whether a continuous variable, known as the *running variable* (X), crosses a specific cutoff or threshold (c). These types of treatment assignment rules appear frequently in policy, education, and healthcare.

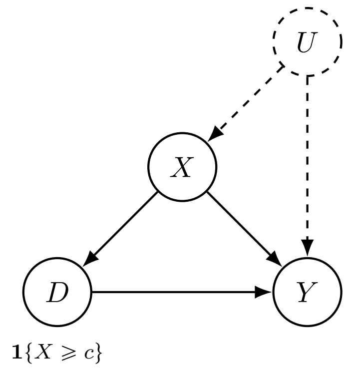
Examples of RDD Settings

- **Minimum Drinking Age (Carpenter & Dobkin, 2009):** In the United States, the legal right to purchase and consume alcohol (D) is granted at age 21 (c). The running variable X is age, and the treatment is assigned via the deterministic rule $D = \mathbf{1}\{\text{Age} \geqslant 21\}$. We can use this discontinuity in the treatment assignment to understand the effect of legal access to alcohol on mortality.

- **Newborn ICU Care (Almond et al., 2010):** In many hospitals, admission to the Neonatal Intensive Care Unit (NICU) (D) is recommended for newborns whose birth weight (X) is below a certain threshold, such as 1500 grams (c). The rule is $D = \mathbf{1}\{\text{Birth Weight} < 1500\text{g}\}$. We can use this discontinuity in the treatment assignment to understand the effect of NICU care on infant mortality.

3.2 The RDD Graph and the Identification Challenge

Let us draw the causal graph for the Sharp RDD setting. We have many unobserved confounders (U) that could be correlated with the running variable. For example, in the newborn case, genetic factors could influence birth weight. However, the running variable X fully determines the treatment D via the cutoff rule, and both X and U can affect the outcome Y.



Discussion: Conditional Exogeneity and Overlap

Question: "Does conditional exogeneity hold in this graph?"

Answer: We can condition on X. If we condition on X, we will block all of the backdoor paths between the treatment assignment and the potential outcome $Y(d)$.

The Catch! So we have conditional exogeneity. But what we do *not* have is **overlap**. For any given value of the running variable X, all individuals are either treated or not treated. The propensity score is always either 0 or 1 if we condition on X.

The Idea: We should be zooming in on the units *around* the threshold. In the small region around the discontinuity, we have both treated and control people. Strictly speaking, the propensity score is always either 0 or 1 if we control for X, but the key insight of RDD is to exploit the *continuity* of the potential outcomes at the cutoff. People slightly with X slightly above c are very similar in terms of their potential outcomes with people with X slightly below c.

3.3 The Smoothness Assumption

The solution to the no-overlap problem is to introduce a new functional form assumption: the **smoothness assumption**. This assumption states that the potential outcomes, as functions of the running variable, are continuous at the cutoff.

The Smoothness Assumption

The conditional expectations of the potential outcomes are continuous functions of the running variable x at the cutoff c. Formally, for $d \in \{0, 1\}$:

$$\mathbb{E}[Y(d) \mid X = x] \text{ is continuous in } x \text{ at } c$$

Intuitively, this means that individuals with X just above the cutoff $(c + \epsilon)$ and just below the cutoff $(c - \epsilon)$ are very similar in all respects, both observed and unobserved, which relate to the outcome of interest, *except for the treatment itself.* As $\epsilon \to 0$, these two groups become effectively interchangeable.

For instance, in the infant mortality example, if we look at the average mortality of infants without ICU treatment in a region near the 1500-gram cutoff, the behavior of that survival rate should be continuous. It cannot be that without ICU treatment, an infant whose birth weight is 1501 grams survives with a very high probability, while one whose weight is 1499 grams survives with a very low probability. The infant mortality rate should only change slightly when we wiggle that birth weight around the threshold. Similarly, had we always been treating infants with an ICU, their mortality rate should not change dramatically around that discontinuity.

When Can the Smoothness Assumption Be Violated?

The smoothness assumption can be violated if the discontinuity point does not only change the treatment assignment but also changes many other things simultaneously. For example:

- **Multiple treatments at the same cutoff:** Suppose that at age 21, people become eligible not only for drinking but also for other things (e.g., if the driving license threshold were also 21). Then at the 21-year-old threshold, people gain access to both alcohol and driving. If we measure extra mortality at age 21, we would not know if it is because people start driving or because they are drinking. Two treatments are administered around the same discontinuity.

- **Bundled interventions:** In the infant weight example, if an infant below 1500 grams is not only put into the ICU but also receives many other treatments simply because they are classified as low birth weight, then those extra treatments will also discontinuously change the mortality rate. There is no way to disentangle the effects unless you make strong assumptions like additivity.

- **Gaming the threshold:** If people know what the threshold is, they might manipulate their running variable to cross it. This is hard to imagine for variables like birth weight or age, but for other running variables (e.g., test scores), manipulation might be easier. Such gaming behavior could invalidate the smoothness assumption.

To check for these issues, you can examine whether other observable variables also change discontinuously at the cutoff. If they do, that is a sign that another treatment is being administered at the same threshold.

3.4 Identification in Sharp RDD: Step-by-Step

Once we are willing to believe in the smoothness assumption, we can proceed with identification. The logic is as follows.

Step 1: Below the cutoff. For individuals with $X < c$, we have $D = 0$ (they are all in the control group). By the overlap condition for $D = 0$ below the cutoff, we can write:

$$\mathbb{E}[Y(0) \mid X = x] = \mathbb{E}[Y(0) \mid X = x, D = 0] = \mathbb{E}[Y \mid X = x, D = 0] = \mathbb{E}[Y \mid X = x] \quad \text{for } x < c$$

where the second equality uses consistency ($Y = Y(0)$ when $D = 0$), and the last equality uses the fact that $D = 0$ is implied when $X < c$.

Step 2: Take the limit from below. By the smoothness assumption, as we take the limit from below, we can identify the potential outcome under control *at the cutoff*:

$$\lim_{x \uparrow c} \mathbb{E}[Y \mid X = x] = \mathbb{E}[Y(0) \mid X = c]$$

So we can fit a regression model predicting the outcome from the running variable using only the data below c, and then take the limit of that regression model at c. This gives us the value of the baseline potential outcome at the discontinuity.

Step 3: Above the cutoff. Similarly, for individuals with $X > c$, we have $D = 1$ (they are all treated). By the same logic:

$$\mathbb{E}[Y(1) \mid X = x] = \mathbb{E}[Y \mid X = x] \quad \text{for } x > c$$

Step 4: Take the limit from above.

$$\lim_{x \downarrow c} \mathbb{E}[Y \mid X = x] = \mathbb{E}[Y(1) \mid X = c]$$

Step 5: The treatment effect at the cutoff. The effect at the discontinuity is the difference of these two limits.

The Sharp RDD Estimand

The Average Treatment Effect at the cutoff, τ_{RDD}, is the difference between the two limits:

$$\tau_{RDD} := \mathbb{E}[Y(1) - Y(0) \mid X = x] = \lim_{x \downarrow c} \mathbb{E}[Y \mid X = x] - \lim_{x \uparrow c} \mathbb{E}[Y \mid X = x]$$

This is the size of the **jump** or **discontinuity** in the observed outcome regression function at the cutoff c. Note that we are identifying the effect only for the **marginal group**—individuals at the cutoff. We are not identifying the overall effect in the population.

Visually, we take all of our samples below the discontinuity (the untreated units), fit a predictive model that predicts the outcome from the running variable, and look at the limit of that predictive model at the discontinuity. Then we move to the samples above the discontinuity (the treated units), fit a separate predictive model, and take the limit of this prediction as the running variable approaches the cutoff from above. The difference between these two limits is our estimated treatment effect.

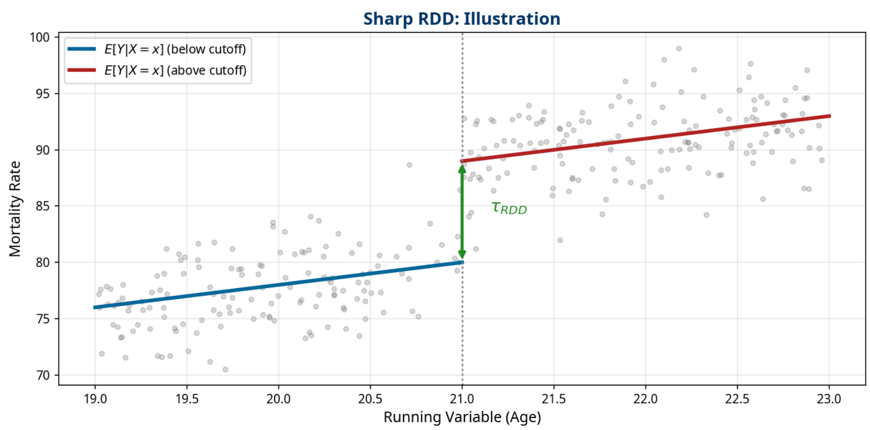


Figure 3: Illustration of the Sharp RDD estimand. The treatment effect is the jump in the regression function at the cutoff.

3.5 Estimation: Local Polynomial Regression

How do we actually estimate the treatment effect in practice? We do not want to make a strong assumption that the underlying model for $\mathbb{E}[Y \mid X = x]$ is linear, because the actual model could be quite complicated. However, we do not really care about the performance of our model across the entire range of the data. We only care about its performance *locally*, right around the discontinuity, because that is where we will estimate the treatment effect.

The standard approach is **local polynomial regression**. We put a kernel, or a weighting function, around the discontinuity. This means we place more weight on samples that are close to the cutoff and less weight on samples that are far away. Then, we learn a simple model—like a linear or a second-degree polynomial model—using these weights. This procedure gives us the best linear (or polynomial) approximation of the more complicated underlying function in the immediate vicinity of the discontinuity.

As we shrink the bandwidth of that kernel closer to zero, the assumption of linearity becomes without loss of generality, because any smooth function can be well-approximated by a line over a small enough interval. So, the standard method is to perform a local linear or polynomial regression with a specific bandwidth, which is equivalent to passing sample weights, determined by a kernel function, into a standard linear regression.

3.6 Bandwidth Selection and Robust Inference

The important parameter in local polynomial regression is the **bandwidth** that you choose. If you choose a uniform kernel with, say, a 2-year window, then you are only using data within 2 years of the cutoff, discarding everything else, and estimating a linear function within those 2 years. The window is the bandwidth. You could also use a Gaussian kernel or another more sophisticated kernel, but every such kernel will still have a bandwidth parameter, and the bandwidth will play a role in the final estimate.

This raises a concern: one could potentially be "p-hacking" based on the choice of bandwidth. Typically, the bandwidth is chosen using cross-validation and mean squared error (MSE) considerations. You choose the bandwidth to optimize for some MSE criterion, but this introduces bias because when optimizing MSE, you are trading off bias and variance.

State-of-the-art methods, developed by Calonico, Cattaneo, and Titiunik (2014), correct for this bias. Their approach both chooses the bandwidth automatically and corrects for the bias introduced by the automatic choice. The `rdrobust` package in R and Python implements these methods.

Python: Sharp RDD with rdrobust

```python
from rdrobust import rdrobust, rdplot
import pandas as pd

# Load data (example from senate elections)
# See: https://rdpackages.github.io/rdrobust/
data = pd.read_csv("rdrobust_senate.csv")

# --- Sharp RDD ---
# y: outcome variable (vote share in next election)
# x: running variable (vote margin at current election)
# c: cutoff (default is 0)
sharp_est = rdrobust(y=data["vote"], x=data["margin"], c=0)

# Print full summary table with:
#   - bias-corrected point estimate
#   - robust confidence intervals
#   - optimal bandwidth (automatically selected)
print(sharp_est)

# --- Visualization ---
# rdplot generates a plot with binned means and
# local polynomial fits on each side of the cutoff
rdplot(y=data["vote"], x=data["margin"], c=0,
       title="Sharp RDD: Incumbency Advantage",
       x_label="Vote Margin",
       y_label="Next Election Vote Share")
```

Note that you can also include extra regressors for variance reduction in the `rdrobust` call, although the primary identification comes from the discontinuity itself.

3.7 Empirical Example: Minimum Drinking Age

Carpenter and Dobkin (2009) used a Sharp RDD to estimate the effect of legal access to alcohol on mortality. The running variable is age, and the cutoff is 21. The researchers fit a local polynomial to the data on either side of the age-21 threshold.

Looking at the difference between the two fitted lines right at the cutoff point, there is a distinct jump: an increase in mortality for individuals around the age of 21, attributable to their gaining legal access to alcohol. The paper found approximately a 9% increase in mortality at the cutoff. The paper further broke this down by presenting similar plots for various types of death. The overall increase in mortality was primarily driven by a rise in traffic accidents—approximately a 15% increase in motor vehicle accident deaths. The data suggests that when young people are legally allowed to drink, there is an increase in drinking and driving, which leads to a higher rate of fatal accidents.

To validate this conclusion, the researchers also examined deaths from internal, disease-related causes. As expected, there was no corresponding effect or jump at the age-21 cutoff. This provides

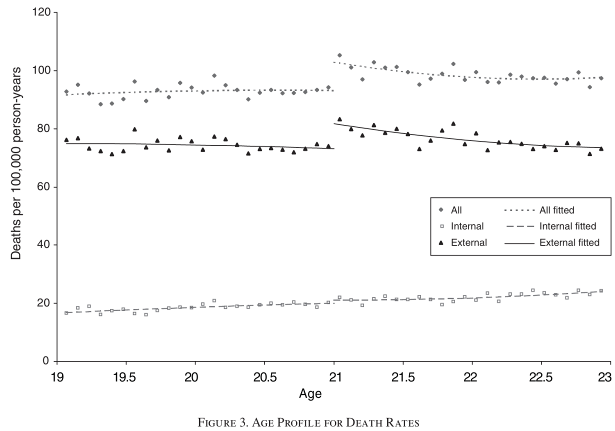


Figure 4: Empirical results from Carpenter and Dobkin (2009).

further evidence that the observed increase in mortality is indeed linked to the change in the legal drinking age, not to some other factor that changes at age 21.

3.8 Empirical Example: At-Risk Newborns

Almond et al. (2010) used a Sharp RDD to estimate the effect of NICU admission on infant mortality. The running variable is birth weight, and the cutoff is 1500 grams. Around that particular birth weight, there is a sharp drop in mortality for infants just below the threshold (who receive ICU care) compared to those just above it (who typically do not). The only thing that changed is a few grams of birth weight, which should not lead to such a large change in mortality on its own. The drop in mortality was because those infants were admitted to the ICU.

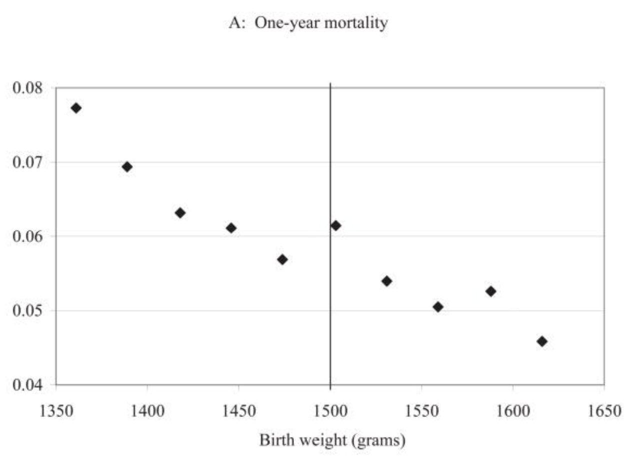


Figure 5: Empirical results from Almond et al. (2010).

The effect is substantial: approximately an 18% relative reduction in infant mortality, with an estimated cost of approximately $550,000 per life saved. As a side comment, if policies were optimal (ignoring treatment costs), at the discontinuity we should see zero effect, because the threshold would be chosen exactly at the point where the benefits and costs cancel out. So, ideally, when we have discontinuous policies, we should see zero effects; otherwise, we should change the policy threshold to better optimize it (or at least to better balance costs and benefits). The fact that we observe a large effect at the 1500-gram threshold suggests that the policy could be improved by shifting the threshold upward.

4 Fuzzy Regression Discontinuity

In many real-world scenarios, the discontinuity at the cutoff is not absolute. In the infant birth weight example, it might not be strictly the case that all infants below the threshold were admitted to the ICU and all infants above it were not. Other factors might lead to an infant with a higher birth weight being admitted (e.g., competing risks or other medical conditions), while capacity constraints at the hospital might mean that some infants below the threshold are not admitted.

This idea of a "fuzzy" discontinuity appears in many policy scenarios:

- **Financial Aid (Van der Klaauw, 2002):** There is typically a sharp eligibility threshold for financial aid based on some academic criteria. Below that threshold, you cannot apply; above it, you can. However, not all individuals who are eligible will actually receive aid. So while everyone below the threshold is not a recipient, not everyone above it is.
- **Medicare (Card, Dobkin, & Maestas, 2009):** Age 65 is a hard threshold for Medicare eligibility. If you are below 65, you are not eligible; if you are 65 or older, you are. This creates a discontinuity. However, not every person over 65 who is eligible for Medicare will actually enroll in the program.

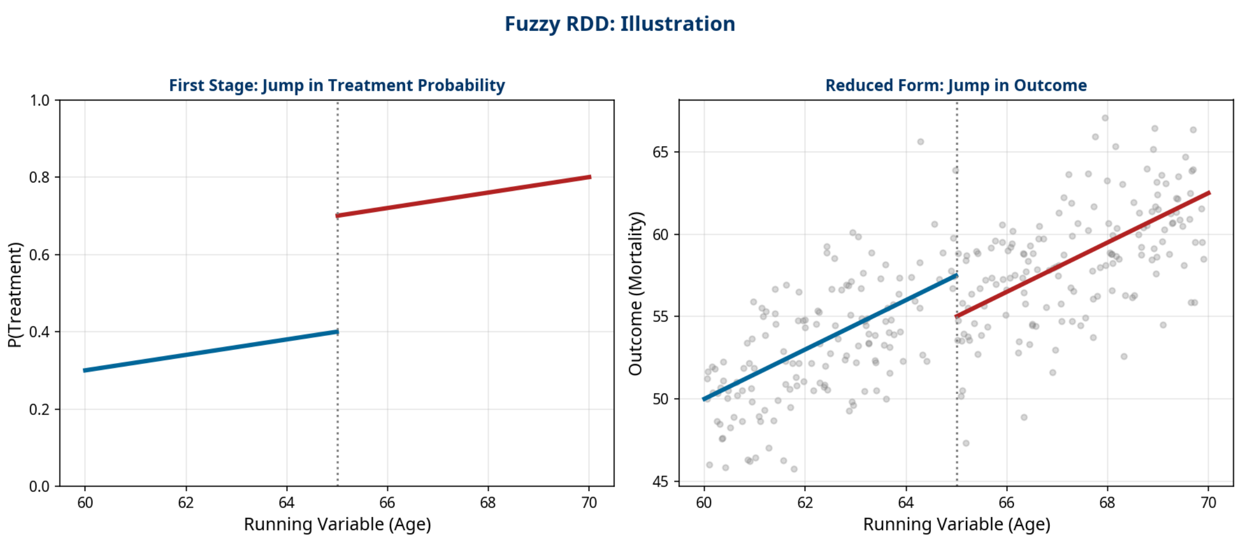
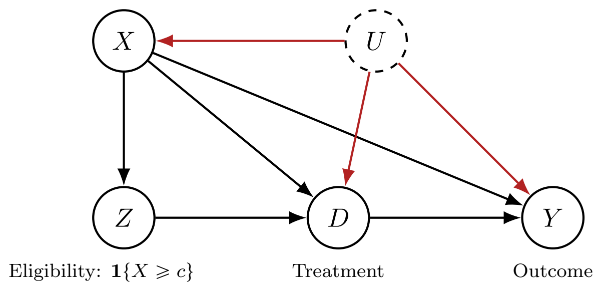


Figure 6: Illustration of a Fuzzy RDD. The cutoff creates a jump in the probability of treatment (First Stage), which in turn creates a jump in the outcome (Reduced Form).

In these cases, the cutoff does not deterministically assign treatment. Instead, it creates a *discontinuity in the probability* of receiving treatment. The probability of treatment jumps from one value to another at the cutoff, but does not go from 0 to 1. This is known as a **Fuzzy Regression Discontinuity Design**.

4.1 The Fuzzy RDD Causal Graph

Let us draw the causal graph for the Fuzzy RDD. We have unobserved confounders U that can affect whether you get treated or not, both above and below the discontinuity. We also have the running variable X, and we can represent the discontinuity around the cutoff with a variable $Z = \mathbf{1}\{X \geqslant c\}$. Whether you are above or below the threshold (Z) will influence your assigned treatment D, but there are also unobserved factors U that determine your treatment.



Discussion: Recognizing the IV Structure

Observation: "This graph looks like an instrumental variable setup!"
Exactly! This is the instrumental variable (IV) graph. Here, Z is our instrument, and D is our treatment, which is confounded. This is a *conditional* instrument, meaning we need to condition on X for Z to be a valid instrument. Since this is an IV graph, we can apply all of the IV analysis that we have learned. Specifically, we can estimate the Local Average Treatment Effect (LATE).

In a Fuzzy RDD, we are essentially running an IV analysis in a narrow window around the cutoff:

- **Instrument:** $Z = \mathbf{1}\{X \geqslant c\}$, the eligibility rule itself.
- **Treatment:** D, the actual treatment received (e.g., enrolling in Medicare, receiving financial aid).
- **Outcome:** Y.

The RDD smoothness assumption ensures that the IV validity conditions (relevance and exclusion) hold *locally* at the cutoff. The exclusion restriction is plausible because the eligibility rule itself should not affect the outcome except through its effect on treatment uptake.

4.2 The Fuzzy RDD Estimand: The Wald Estimator

Since Fuzzy RDD is a local IV, the causal estimand is the Local Average Treatment Effect (LATE) for compliers. However, we face the same overlap problem as in the Sharp RDD: for any given value of X, the instrument Z is deterministically either 0 or 1. So we cannot write the standard IV formula directly. However, we *can* write it near the discontinuity. To identify the LATE at the cutoff, we need to estimate two things:

1. The effect of the instrument Z on the **outcome** Y, conditional on X being at the discontinuity. This is estimated exactly like a Sharp RDD for the outcome.

2. The effect of the instrument Z on the **treatment** D, conditional on X being at the discontinuity. This is estimated exactly like a Sharp RDD for the treatment.

The Fuzzy RDD Estimand (Wald Estimator)

The LATE for compliers at the cutoff is:

$$\begin{aligned}\tau_{\text{Fuzzy}} = \mathbb{E}[Y(1) - Y(0) \mid X = c, \text{complier}] &= \frac{\lim_{x \downarrow c} \mathbb{E}[Y|X = x] - \lim_{x \uparrow c} \mathbb{E}[Y|X = x]}{\lim_{x \downarrow c} \mathbb{E}[D|X = x] - \lim_{x \uparrow c} \mathbb{E}[D|X = x]} \\ &= \frac{\text{Jump in Outcome}}{\text{Jump in Treatment Probability}}\end{aligned}$$

This is the Wald estimator applied to the discontinuities. The compliers in this setting are the individuals at the cutoff whose treatment status is changed by crossing the threshold.

Discussion: Who Are the Compliers?

Question: “What is an example of a complier in this context?”
Response: A complier in the newborn example would be an infant who, at a birth weight of 1499 grams, would be admitted to the ICU, but at 1501 grams, would *not* be admitted to the ICU. This assumes no capacity constraints at the hospital and no other competing factors—complete compliance at the discontinuity.
Question: “If there is complete compliance, what happens to the denominator?”
Response: If compliance is perfect (everyone above the cutoff is treated, everyone below is not), then the denominator—the jump in treatment probability—is simply 1. In that case, the Fuzzy RDD reduces to the Sharp RDD. The Sharp RDD is a special case of the Fuzzy RDD where the first stage is a step function.

The `rdrobust` package handles Fuzzy RDD estimation seamlessly by simply adding the `fuzzy` parameter to specify the treatment variable. The package automatically computes the Wald estimator.

Python: Fuzzy RDD with `rdrobust`

```python
from rdrobust import rdrobust
import pandas as pd

# --- Fuzzy RDD ---
# fuzzy: the endogenous treatment variable
# The package automatically calculates the Wald estimator:
#   (jump in outcome) / (jump in treatment probability)

# Example: effect of financial aid on college enrollment
# Y: college_enrollment (outcome)
# X: academic_index    (running variable)
# D: received_aid      (actual treatment, imperfect compliance)
c = 75  # Cutoff for aid eligibility

fuzzy_est = rdrobust(y=data["college_enrollment"],
                     x=data["academic_index"],
                     c=c,
                     fuzzy=data["received_aid"])

# Print summary with LATE estimate and robust CIs
print(fuzzy_est)
```

5 Activity: Spot the Design

To consolidate the material from this lecture, let's work through three scenarios. For each scenario, we discuss: (1) Is this a DiD, Sharp RDD, or Fuzzy RDD problem? (2) What are the key variables? (3) What is the key identifying assumption?

5.1 Scenario 1: State-Level Minimum Wage Increase

Problem: A state raises its minimum wage. To estimate the effect on employment, an economist collects monthly employment data from this state and a neighboring state that did not change its wage.

Answer: Difference-in-Differences

- **Design:** Difference-in-Differences (DiD).
- **Treatment:** Minimum wage increase.
- **Outcome:** Employment level.
- **Key Assumption:** Parallel trends—employment growth would have been the same in both states absent the wage increase.

This is the classic Card & Krueger (1994) setup, which compared employment in fast-food restaurants in New Jersey (which raised its minimum wage) and Pennsylvania (which did not).

5.2 Scenario 2: University Scholarship Program

Problem: A university offers a merit scholarship to all applicants with an SAT score of 1400 or higher. We want to know the effect of the scholarship on enrollment.

Answer: Sharp RDD

- **Design:** Sharp Regression Discontinuity (Sharp RDD).
- **Treatment:** Receiving the scholarship.
- **Outcome:** Enrollment.
- **Running Variable:** SAT score, with cutoff at 1400.
- **Key Assumption:** Smoothness of potential outcomes at the 1400 SAT cutoff.

Since the scholarship is deterministically assigned based on the SAT score cutoff, this is a Sharp RDD.

5.3 Scenario 3: Get-Out-the-Vote Campaign

Problem: A campaign sends mailers to all registered voters in households with an average age of 50 or older. We want to know the effect of the mailer on voting, but we know some younger households might receive one by mistake and some older households might not.

Answer: Fuzzy RDD

- **Design:** Fuzzy Regression Discontinuity (Fuzzy RDD).
- **Treatment:** Receiving the mailer.
- **Outcome:** Voting.
- **Running Variable:** Average household age, with cutoff at 50.
- **Key Assumption:** Smoothness of potential outcomes at the age-50 cutoff.

Because compliance is imperfect—some younger households receive the mailer and some older households do not—this is a Fuzzy RDD. The eligibility rule (age $\geqslant 50$) serves as the instrument, and the actual receipt of the mailer is the treatment.

6 Conclusion and Key Takeaways

This chapter introduced two powerful and widely used strategies for causal inference in the presence of unobserved confounding: Difference-in-Differences and Regression Discontinuity. Unlike the methods we studied in previous lectures, their identifying power comes not from graphical structure alone, but from assumptions about the functional form of the underlying relationships.

1. **Beyond DAGs:** DiD and RDD rely on **functional form assumptions** (parallel trends and smoothness) that cannot be fully represented in a causal DAG. This is qualitatively different from the front-door criterion, instrumental variables, and proximal inference, which are all identified from the graph structure.
2. **Difference-in-Differences (DiD):** Compares the *change* in outcomes over time between treated and control groups. It is identified by the **parallel trends assumption**. This assumption can be made more plausible by conditioning on covariates (Conditional DiD) and can be partially checked with placebo tests. The Doubly Robust estimator for the ATT provides a robust way to estimate Conditional DiD models with machine learning.
3. **Regression Discontinuity (RDD):** Exploits a cutoff in a running variable to create a local randomized experiment. **Sharp RDD** assumes perfect compliance at the cutoff, while **Fuzzy RDD** handles imperfect compliance and is identified as a local instrumental variable design.
4. **Local Effects:** A key feature of these methods is that they identify *local* effects. RDD identifies the treatment effect only for the population at the cutoff, which may not generalize to the broader population. DiD identifies the Average Treatment Effect on the Treated (ATT) for the treated population.

Both DiD and RDD are mainstays of the applied econometrician's toolkit. Their validity, however, hinges on careful consideration of their underlying assumptions in the context of the specific research question.

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